

## Task : Bug Fixes

### 1. SQL precedence Bug (TaskRepository.java)

problem:

- AND bound tighter than OR in SQL
- So the status and archived filters were applied only to part of search condition.

Before

WHERE title LIKE ? OR description LIKE ?  
 AND status = ? AND archived = ?

- AND applies only to description part.

After

WHERE (title LIKE ? OR description LIKE ?)  
 AND status = ? AND archived = ?

⇒ Added parentheses to search condition in TaskRepository.java

### 2. Search Wildcards

problem:

- % and \_ matched everything because they are SQL wildcards

solution

- Escape the % and \_ characters in controller

String search = searchTerm.replace(
 "%", "\\%").replace(
 "\_", "\\\_");

### 3. Controller (TaskController.java)

- Remove artificial Thread sleep
- Added validation for page 0, bad pageSize and invalid status
- Now returns 400 (Bad Request) with message instead of 500

### 4. Stable Ordering

- Added id as a tie-breaker in sorting
- Sorted in ascending order.

### 5. Race Condition (useTasks.js)

Problem: stale responses could overwrite newer ones.

Fix

- Used AbortController to cancel previous requests
  - Fixed loading & error state so spinner no longer sticks after failure.
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